CSE563 — Applied Mathematics

Practice Question Bank

Unit-wise • numerical only • strictly within the prescribed syllabus

School of Computer Science & Engineering • Session 2026–27

Questions with Answer Key

Contents

Unit I — Basic Mathematics in Computer Science 2

Unit II — Vectors and Matrices 5

Unit III — Probability and Statistics 10

Unit IV — Graph Theory 14

Unit V — Algorithm Complexity Analysis 18

Unit VI — Algorithmic and Optimization Approach 22

Unit I — Basic Mathematics in Computer Science

Maps to CO1 — recall the basic concepts of mathematics in computer science and engineering.

Syllabus: number theory, set theory, boolean algebra, logic and calculus, mathematical induction, functions and relations.

This unit: 4 short-answer questions (2 marks) and 4 long-answer questions (10 marks) = **48 marks**.

Part A — Short Answer Questions — 2 marks each

Q1. *[2 marks]* State the value of $\gcd(72, 30)$ using the Euclidean algorithm, and hence find $\text{lcm}(72, 30)$.

Syllabus: Number theory • Euclidean algorithm, GCD and LCM • CO1 • Bloom L1 (Remember) • Easy • Analytical • Start word: State

Solution. $72 = 2(30) + 12$; $30 = 2(12) + 6$; $12 = 2(6) + 0$. So gcd = **6**.

$$\text{lcm}(72, 30) = \frac{72 \times 30}{6} = \frac{2160}{6} = \mathbf{360}$$

Q2. *[2 marks]* Illustrate the inclusion–exclusion principle by finding how many integers from 1 to 200 are divisible by 3 or by 4.

Syllabus: Set theory • Inclusion–exclusion principle • CO1 • Bloom L2 (Understand) • Easy • Analytical • Start word: Illustrate

Solution. $|A| = \lfloor 200/3 \rfloor = 66$, $|B| = \lfloor 200/4 \rfloor = 50$, $|A \cap B| = \lfloor 200/12 \rfloor = 16$.

$$|A \cup B| = 66 + 50 - 16 = \mathbf{100}$$

Q3. *[2 marks]* Calculate the truth value of $(p \vee q) \to r$ for all combinations of p, q, r, and state in how many rows the expression is true.

Syllabus: Logic • Truth table of an implication • CO1 • Bloom L3 (Apply) • Medium • Analytical • Start word: Calculate

Solution.

p	q	r	$p \vee q$	$(p \vee q) \to r$
T	T	T	T	T
T	T	F	T	**F**
T	F	T	T	T
T	F	F	T	**F**
F	T	T	T	T
F	T	F	T	**F**
F	F	T	F	T
F	F	F	F	T

True in **5** of the 8 rows; false in the three rows where $p \vee q = \text{T}$ and $r = \text{F}$.

Q4. *[2 marks]* Determine the number of equivalence classes of congruence modulo 5 on $\mathbb{Z}$, and list the four smallest non-negative members of the class containing 12.

Syllabus: Functions and relations • Equivalence classes of congruence • CO1 • Bloom L3 (Apply) • Easy • Analytical • Start word: Determine

Solution. Congruence modulo 5 gives **5** classes: $[0], [1], [2], [3], [4]$.
$12 \bmod 5 = 2$, so $12 \in [2]$. Its four smallest non-negative members are **2**, **7**, **12**, **17**.

Part B — Long Answer Questions — 10 marks each

Q5. *[10 marks]* Apply the rules of calculus.

(a) Find $\dfrac{dy}{dx}$ for $y = 4x^3 - 6x^2 + 5$.

(b) For $f(x, y) = x^3y + 2xy$, find $\dfrac{\partial f}{\partial x}$ and $\dfrac{\partial f}{\partial y}$.

(c) Evaluate both partial derivatives at the point $(1, 2)$.

(d) Evaluate $\displaystyle\int_1^4 (2x + 3)\, dx$.

Syllabus: Calculus • Derivatives, partial derivatives and definite integral • CO1 • Bloom L3 (Apply) • Medium • Analytical • Start word: Apply

Solution. **(a)** $\dfrac{dy}{dx} = \mathbf{12x^2 - 12x}$.
(b) $\dfrac{\partial f}{\partial x} = \mathbf{3x^2y + 2y}$; $\dfrac{\partial f}{\partial y} = \mathbf{x^3 + 2x}$.
(c) At $(1, 2)$: $f_x = 3(1)(2) + 2(2) = \mathbf{10}$; $f_y = 1^3 + 2(1) = \mathbf{3}$.
(d) $\displaystyle\int_1^4 (2x + 3)\, dx = \big[x^2 + 3x\big]_1^4 = (16 + 12) - (1 + 3) = 28 - 4 = \mathbf{24}$.

Q6. *[10 marks]* Examine arithmetic modulo 11.

(a) Compute $4^3 \bmod 11$.

(b) Compute $6^2 \bmod 11$.

(c) Find the remainder when 2^{10} is divided by 11.

(d) Find the value of x in $\{0, 1, \ldots, 10\}$ for which $5x \equiv 1 \pmod{11}$.

Syllabus: Number theory • Modular arithmetic • CO1 • Bloom L4 (Analyse) • Medium • Analytical • Start word: Examine

Solution. **(a)** $4^3 = 64 = 5(11) + 9$, so $4^3 \equiv \mathbf{9} \pmod{11}$.
(b) $6^2 = 36 = 3(11) + 3$, so $6^2 \equiv \mathbf{3} \pmod{11}$.
(c) $2^{10} = 1024 = 93(11) + 1$, so the remainder is **1**. (Shorter: 11 is prime and $11 \nmid 2$, so $2^{10} \equiv 1$ by Fermat's little theorem.)

(d) Test the multiples of 5: $5(9) = 45 = 4(11) + 1$, so $x = \mathbf{9}$.

Q7. *[10 marks]* Evaluate $F(A, B, C) = \sum m(0, 2, 4, 5, 6, 7)$ using a three-variable Karnaugh map.

(a) Write the six minterms in binary as ABC.

(b) Plot the map and mark the two largest groups.

(c) State the minimal sum-of-products expression.

(d) Count the literals in the canonical and minimal forms and state the saving.

Syllabus: Boolean algebra • Karnaugh map minimisation • CO1 • Bloom L5 (Evaluate) • Difficult • Analytical • Start word: Evaluate

Solution. **(a)** $m_0 = 000$, $m_2 = 010$, $m_4 = 100$, $m_5 = 101$, $m_6 = 110$, $m_7 = 111$.

(b)

$A\backslash BC$	00	01	11	10
0	**1**	0	0	**1**
1	**1**	**1**	**1**	**1**

Group 1: the whole row $A = 1$ (four cells m_4, m_5, m_7, m_6) $\Rightarrow A$. Group 2: the two columns with $C = 0$ (four cells m_0, m_2, m_4, m_6) $\Rightarrow \overline{C}$.

(c) $\mathbf{F = A + \overline{C}}$.

(d) Canonical form: 6 minterms $\times$ 3 literals = **18** literals. Minimal form: **2** literals. Saving = **16** literals.

Q8. *[10 marks]* Construct a numerical verification of the formula $1^3 + 2^3 + \cdots + n^3 = \left[\dfrac{n(n+1)}{2}\right]^2$.

(a) Verify by direct addition for $n = 1, 2, 3, 4, 5$.

(b) Compute both sides for $n = 6$ and $n = 7$.

(c) Evaluate the formula for $n = 10$ and $n = 100$.

(d) Find the value of n for which the sum equals 1296.

Syllabus: Mathematical induction • Verification of the sum formula • CO1 • Bloom L6 (Create) • Difficult • Analytical • Start word: Construct

Solution. **(a)**

n	Sum of cubes	$n(n+1)/2$	Square
1	1	1	1
2	9	3	9
3	36	6	36
4	100	10	100
5	225	15	225

(b) $n = 6$: sum $= 225 + 216 = \mathbf{441}$ and $(21)^2 = 441$. $n = 7$: sum $= 441 + 343 = \mathbf{784}$ and $(28)^2 = 784$.

(c) $n = 10$: $(55)^2 = \mathbf{3025}$. $n = 100$: $(5050)^2 = \mathbf{25\,502\,500}$.

(d) $\left[\frac{n(n+1)}{2}\right]^2 = 1296 \Rightarrow \frac{n(n+1)}{2} = 36 \Rightarrow n(n+1) = 72 \Rightarrow n = \mathbf{8}$.

Unit II — Vectors and Matrices

Maps to CO2 — demonstrate the use of vectors and matrices in image transformations.

Syllabus: vector and operations, matrices and operations, principal component analysis, linear discriminant analysis, image transformation, image interpolation.

This unit: 8 short-answer questions (2 marks) and 4 long-answer questions (10 marks) = **56 marks**.

Part A — Short Answer Questions — 2 marks each

Q9. *[2 marks]* State the value of $\mathbf{u} \cdot \mathbf{v}$ for $\mathbf{u} = (2, -3, 4)$ and $\mathbf{v} = (5, 1, -2)$.

Syllabus: Vector and operations • Dot product • CO2 • Bloom L1 (Remember) • Easy • Analytical • Start word: State

Solution.

$$\mathbf{u} \cdot \mathbf{v} = 2(5) + (-3)(1) + 4(-2) = 10 - 3 - 8 = \mathbf{-1}$$

Q10. *[2 marks]* State the Euclidean norm of $\mathbf{w} = (2, 3, 6)$, and the Euclidean distance between the points $P(2, 1)$ and $Q(7, 13)$.

Syllabus: Vector and operations • Euclidean norm and Euclidean distance • CO2 • Bloom L1 (Remember) • Easy • Analytical • Start word: State

Solution.

$$\|\mathbf{w}\| = \sqrt{4 + 9 + 36} = \sqrt{49} = \mathbf{7}$$

$$d(P, Q) = \sqrt{(7-2)^2 + (13-1)^2} = \sqrt{25 + 144} = \sqrt{169} = \mathbf{13}$$

Q11. *[2 marks]* Illustrate the cross product by evaluating $(3, 1, 2) \times (1, 4, 2)$.

Syllabus: Vector and operations • Cross product • CO2 • Bloom L2 (Understand) • Medium • Analytical • Start word: Illustrate

Solution.

$$(3, 1, 2) \times (1, 4, 2) = (1(2) - 2(4),\ 2(1) - 3(2),\ 3(4) - 1(1)) = (\mathbf{-6},\ \mathbf{-4},\ \mathbf{11})$$

Check: the result is orthogonal to both inputs — $(-6, -4, 11) \cdot (3, 1, 2) = -18 - 4 + 22 = 0$. ✓

Q12. *[2 marks]* Illustrate the evaluation of a 2×2 determinant by finding $\det(A)$ and the inverse for $A = \begin{pmatrix} 5 & 2 \\ 3 & 4 \end{pmatrix}$.

Syllabus: Matrices and operations • Determinant and inverse • CO2 • Bloom L2 (Understand) • Easy • Analytical • Start word: Illustrate

Solution.

$$\det A = 5(4) - 2(3) = 20 - 6 = \mathbf{14}$$

$$A^{-1} = \frac{1}{14}\begin{pmatrix} 4 & -2 \\ -3 & 5 \end{pmatrix} = \begin{pmatrix} 2/7 & -1/7 \\ -3/14 & 5/14 \end{pmatrix}$$

Q13. *[2 marks]* Determine the rank of $\begin{pmatrix} 1 & 2 & 3 \\ 2 & 4 & 7 \end{pmatrix}$ and the dimension of its null space.

Syllabus: Matrices and operations • Rank and nullity • CO2 • Bloom L3 (Apply) • Easy • Analytical • Start word: Determine

Solution. $R_2 - 2R_1 = (0, 0, 1)$, which is non-zero, so both rows are independent and the **rank = 2**.

By the rank–nullity theorem with 3 columns, **nullity** $= 3 - 2 = \mathbf{1}$.

Q14. *[2 marks]* Calculate the eigenvalues of $A = \begin{pmatrix} 5 & 2 \\ 1 & 4 \end{pmatrix}$.

Syllabus: Matrices and operations • Eigenvalues • CO2 • Bloom L3 (Apply) • Medium • Analytical • Start word: Calculate

Solution. $\operatorname{tr} A = 9$ and $\det A = 20 - 2 = 18$, so

$$\lambda^2 - 9\lambda + 18 = 0 \;\Rightarrow\; (\lambda - 6)(\lambda - 3) = 0 \;\Rightarrow\; \lambda = \mathbf{6},\ \mathbf{3}$$

Check: $6 + 3 = 9 = \operatorname{tr} A$ and $6 \times 3 = 18 = \det A$. ✓

Q15. *[2 marks]* Compute the pixel value at (0.6, 0.3) by bilinear interpolation, given $f(0,0) = 20$, $f(1,0) = 60$, $f(0,1) = 40$, $f(1,1) = 80$. State also the nearest-neighbour value.

Syllabus: Image interpolation • Bilinear and nearest-neighbour interpolation • CO2 • Bloom L3 (Apply) • Medium • Analytical • Start word: Compute

Solution. With $a = 0.6$ and $b = 0.3$:

$$f = (0.4)(0.7)(20) + (0.6)(0.7)(60) + (0.4)(0.3)(40) + (0.6)(0.3)(80)$$

$$= 5.6 + 25.2 + 4.8 + 14.4 = \mathbf{50.0}$$

The four weights are $0.28, 0.42, 0.12, 0.18$ and sum to 1. ✓

Nearest neighbour: $(0.6, 0.3) \to (1, 0)$, giving **60**.

Q16. *[2 marks]* Analyze a PCA result whose covariance matrix has eigenvalues 8, 6, 4 and 2. Compute the percentage of variance explained by PC1, and by PC1 and PC2 together.

Syllabus: Principal component analysis • Variance explained • CO2 • Bloom L4 (Analyse) • Medium • Analytical • Start word: Analyze

Solution. Total variance $= 8 + 6 + 4 + 2 = 20$.

$$\text{PC1} = \frac{8}{20} = \mathbf{40}\%, \qquad \text{PC1} + \text{PC2} = \frac{8+6}{20} = \frac{14}{20} = \mathbf{70}\%$$

Part B — Long Answer Questions — 10 marks each

Q17. *[10 marks]* Solve the system $x + y + z = 4$, $2y + 3z = 7$, $3x + 2y - z = 1$ by Gaussian elimination.

(a) Write the augmented matrix.

(b) Reduce it to row echelon form.

(c) Back-substitute for x, y and z.

(d) Substitute your solution back into all three equations.

Syllabus: Matrices and operations • Gaussian elimination • CO2 • Bloom L3 (Apply) • Medium • Analytical • Start word: Solve

Solution. **(a)** $\left(\begin{array}{ccc|c} 1 & 1 & 1 & 4 \\ 0 & 2 & 3 & 7 \\ 3 & 2 & -1 & 1 \end{array}\right)$

(b) $R_3 - 3R_1$ gives $(0, -1, -4 \mid -11)$; then $2R_3 + R_2$ gives $(0, 0, -5 \mid -15)$:

$$\left(\begin{array}{ccc|c} 1 & 1 & 1 & 4 \\ 0 & 2 & 3 & 7 \\ 0 & 0 & -5 & -15 \end{array}\right)$$

(c) $-5z = -15 \Rightarrow z = \mathbf{3}$; $\quad 2y + 3(3) = 7 \Rightarrow y = \mathbf{-1}$; $\quad x = 4 - (-1) - 3 = \mathbf{2}$.

(d) $2 + (-1) + 3 = 4$ ✓; $\quad 2(-1) + 3(3) = 7$ ✓; $\quad 3(2) + 2(-1) - 3 = 6 - 2 - 3 = 1$ ✓.

Q18. *[10 marks]* Examine the data set $(1, 2), (2, 1), (3, 4), (4, 3), (5, 5)$ by principal component analysis.

(a) Compute the mean and centre the data.

(b) Compute the 2×2 covariance matrix (divide by $n - 1$).

(c) Compute both eigenvalues.

(d) Compute the percentage of variance explained by PC1.

(e) Project all five centred points onto $(1, 1)/\sqrt{2}$.

Syllabus: Principal component analysis • Covariance and projection • CO2 • Bloom L4 (Analyse) • Difficult • Analytical • Start word: Examine

Solution. **(a)** $\bar{x} = 3$, $\bar{y} = 3$. Centred: $(-2, -1), (-1, -2), (0, 1), (1, 0), (2, 2)$.

(b) $\sum \tilde{x}^2 = 10$, $\sum \tilde{y}^2 = 10$, $\sum \tilde{x}\tilde{y} = 2 + 2 + 0 + 0 + 4 = 8$.

$$S = \frac{1}{4}\begin{pmatrix} 10 & 8 \\ 8 & 10 \end{pmatrix} = \begin{pmatrix} 2.5 & 2 \\ 2 & 2.5 \end{pmatrix}$$

(c) $\lambda^2 - 5\lambda + 2.25 = 0 \Rightarrow \lambda = \dfrac{5 \pm 4}{2}$, so $\lambda_1 = \mathbf{4.5}$ and $\lambda_2 = \mathbf{0.5}$. (Sum $= 5 = \operatorname{tr} S$. ✓)

(d) PC1 explains $\dfrac{4.5}{5} = \mathbf{90}\%$.

(e) Projection $= (\tilde{x} + \tilde{y})/\sqrt{2}$:

$$\mathbf{-2.1213},\ \mathbf{-2.1213},\ \mathbf{0.7071},\ \mathbf{0.7071},\ \mathbf{2.8284}$$

Q19. *[10 marks]* Evaluate a composite image transformation in which a point is first translated by $(2, -1)$ and then scaled by 2 about the origin.

(a) Write the 3×3 homogeneous matrices T and S.

(b) Compute $M = S\,T$.

(c) Apply M to $(3, 4)$, showing each stage.

(d) Compute $T\,S$ and apply it to $(3, 4)$.

Syllabus: Image transformation • Homogeneous coordinates and composition • CO2 • Bloom L5 (Evaluate) • Difficult • Analytical • Start word: Evaluate

Solution. **(a)** $T = \begin{pmatrix} 1 & 0 & 2 \\ 0 & 1 & -1 \\ 0 & 0 & 1 \end{pmatrix}$, $\quad S = \begin{pmatrix} 2 & 0 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 1 \end{pmatrix}$.

(b) $M = S\,T = \begin{pmatrix} 2 & 0 & 4 \\ 0 & 2 & -2 \\ 0 & 0 & 1 \end{pmatrix}$, i.e. $(x, y) \mapsto (2x + 4,\ 2y - 2)$.

(c) Translate: $(3, 4) \to (5, 3)$. Scale: $(5, 3) \to (\mathbf{10}, \mathbf{6})$. Directly: $(2(3) + 4,\ 2(4) - 2) = (10, 6)$. ✓

(d) $T\,S = \begin{pmatrix} 2 & 0 & 2 \\ 0 & 2 & -1 \\ 0 & 0 & 1 \end{pmatrix}$, giving $(2(3) + 2,\ 2(4) - 1) = (\mathbf{8}, \mathbf{7})$. The two orders differ, because scaling multiplies any translation already applied.

Q20. *[10 marks]* Construct a Fisher linear discriminant for $C_1 = \{(1, 2), (2, 1), (3, 3)\}$ and $C_2 = \{(6, 7), (7, 6), (8, 8)\}$.

(a) Compute both class means.

(b) Compute S_1, S_2 and S_W.

(c) Compute $\mathbf{w} = S_W^{-1}(\mathbf{m}_2 - \mathbf{m}_1)$.

(d) Project all six points onto $\mathbf{w}$ and compute the threshold midway between the projected means.

Syllabus: Linear discriminant analysis • Fisher direction and threshold • CO2 • Bloom L6 (Create) • Difficult • Analytical • Start word: Construct

Solution. **(a)** $\mathbf{m}_1 = (2, 2)$, $\quad \mathbf{m}_2 = (7, 7)$.

(b) C_1 deviations: $(-1, 0), (0, -1), (1, 1)$, so $\sum d_x^2 = 2$, $\sum d_y^2 = 2$, $\sum d_x d_y = 1$:

$$S_1 = \begin{pmatrix} 2 & 1 \\ 1 & 2 \end{pmatrix}$$

C_2 has the identical deviation pattern, so $S_2 = S_1$ and

$$S_W = \begin{pmatrix} 4 & 2 \\ 2 & 4 \end{pmatrix}, \qquad S_W^{-1} = \frac{1}{12}\begin{pmatrix} 4 & -2 \\ -2 & 4 \end{pmatrix}$$

(c) $\mathbf{m}_2 - \mathbf{m}_1 = (5, 5)$, so

$$\mathbf{w} = \frac{1}{12}\begin{pmatrix} 4 & -2 \\ -2 & 4 \end{pmatrix}\begin{pmatrix} 5 \\ 5 \end{pmatrix} = \begin{pmatrix} 5/6 \\ 5/6 \end{pmatrix} \propto (\mathbf{1}, \mathbf{1})$$

(d) Using $\mathbf{w} = (1, 1)$, the projections are $x + y$:

$$C_1 : \ 3,\ 3,\ 6 \text{ (mean 4)}; \qquad C_2 : \ 13,\ 13,\ 16 \text{ (mean 14)}$$

$$\text{threshold} = \frac{4 + 14}{2} = \mathbf{9}$$

Every C_1 projection is below 9 and every C_2 projection above it, so the two classes separate perfectly.

Unit III — Probability and Statistics

Maps to CO3 — apply the concepts of probability and statistics for inference and regression analysis.

Syllabus: basic probability, compound probability, probability distribution, frequentist inference, bayesian inference, regression analysis.

This unit: 8 short-answer questions (2 marks) and 4 long-answer questions (10 marks) = **56 marks**.

Part A — Short Answer Questions — 2 marks each

Q21. *[2 marks]* State the probability of getting a sum of 6 when two fair dice are rolled, listing the favourable outcomes.

Syllabus: Basic probability • Sample space • CO3 • Bloom L1 (Remember) • Easy • Analytical • Start word: State

Solution. Favourable outcomes: $(1,5),(2,4),(3,3),(4,2),(5,1)$ — that is 5 of the 36 equally likely outcomes.

$$P(\text{sum}=6)=\frac{5}{36}\approx \mathbf{0.1389}$$

Q22. *[2 marks]* State the mean, variance and standard deviation of a binomial variable with $n=20$ and $p=0.3$.

Syllabus: Probability distribution • Binomial mean and variance • CO3 • Bloom L1 (Remember) • Easy • Analytical • Start word: State

Solution.

$$\mu=np=20(0.3)=\mathbf{6},\qquad \sigma^2=np(1-p)=20(0.3)(0.7)=\mathbf{4.2}$$

$$\sigma=\sqrt{4.2}\approx \mathbf{2.0494}$$

Q23. *[2 marks]* Illustrate the Poisson distribution by computing $P(X=2)$ when $\lambda=4$, correct to four decimal places.

Syllabus: Probability distribution • Poisson distribution • CO3 • Bloom L2 (Understand) • Medium • Analytical • Start word: Illustrate

Solution.

$$P(X=2)=\frac{e^{-4}\,4^2}{2!}=\frac{0.0183156\times 16}{2}=0.0183156\times 8=\mathbf{0.1465}$$

Q24. *[2 marks]* Illustrate the addition rule by computing $P(A\cup B)$ when $P(A)=0.5$, $P(B)=0.4$ and $P(A\cap B)=0.2$. Compute $P(A)P(B)$ and compare it with $P(A\cap B)$.

Syllabus: Basic probability • Addition rule • CO3 • Bloom L2 (Understand) • Easy • Analytical • Start word: Illustrate

Solution.

$$P(A \cup B) = 0.5 + 0.4 - 0.2 = \mathbf{0.7}$$

$P(A)P(B) = 0.5 \times 0.4 = \mathbf{0.20}$, which equals $P(A \cap B) = 0.20$, so here the two events are **independent**.

Q25. *[2 marks]* Compute $P(A \mid B)$ and $P(B \mid A)$ when $P(A) = 0.5$, $P(B) = 0.4$ and $P(A \cap B) = 0.2$.

Syllabus: Compound probability • Conditional probability • CO3 • Bloom L3 (Apply) • Medium • Analytical • Start word: Compute

Solution.

$$P(A \mid B) = \frac{0.2}{0.4} = \mathbf{0.5}, \qquad P(B \mid A) = \frac{0.2}{0.5} = \mathbf{0.4}$$

Each conditional equals the corresponding unconditional probability, which is the signature of independence.

Q26. *[2 marks]* Calculate $P(X > 85)$ for $X \sim N(70,\, 10^2)$, stating the z-score used.

Syllabus: Probability distribution • Normal distribution • CO3 • Bloom L3 (Apply) • Medium • Analytical • Start word: Calculate

Solution.

$$z = \frac{85 - 70}{10} = \mathbf{1.5}$$

$$P(X > 85) = 1 - \Phi(1.5) = 1 - 0.9332 = \mathbf{0.0668} \quad (6.68\%)$$

Q27. *[2 marks]* Determine the standard error of the mean and the half-width of a 95% confidence interval for a sample of $n = 100$ with $s = 25$.

Syllabus: Frequentist inference • Standard error and confidence interval • CO3 • Bloom L3 (Apply) • Medium • Analytical • Start word: Determine

Solution.

$$\text{SE} = \frac{s}{\sqrt{n}} = \frac{25}{10} = \mathbf{2.5}$$

$$\text{half-width} = 1.96 \times 2.5 = \mathbf{4.9}$$

Q28. *[2 marks]* Analyze two regression models with correlation coefficients $r = 0.5$ and $r = 0.8$. Compute R^2 for each and the difference in the percentage of variance explained.

Syllabus: Regression analysis • Coefficient of determination • CO3 • Bloom L4 (Analyse) • Medium • Analytical • Start word: Analyze

Solution.

$$R^2 = 0.5^2 = \mathbf{0.25}\ (25\%), \qquad R^2 = 0.8^2 = \mathbf{0.64}\ (64\%)$$

Difference $= 64 - 25 = \mathbf{39}$ percentage points. Raising r by 0.3 more than doubles the variance explained.

Part B — Long Answer Questions — 10 marks each

Q29. *[10 marks]* Analyze the following production problem. Machine A makes 70% of items with 3% defective; machine B makes 30% with 8% defective. One item is chosen at random.

(a) Compute $P(A \cap D)$ and $P(B \cap D)$.

(b) Compute $P(D)$.

(c) Compute $P(A \mid D)$.

(d) Compute $P(B \mid D)$ and check that the two posteriors sum to 1.

Syllabus: Compound probability • Bayes' theorem • CO3 • Bloom L4 (Analyse) • Difficult • Analytical • Start word: Analyze

Solution. **(a)** $P(A \cap D) = 0.70(0.03) = \mathbf{0.021}$; $P(B \cap D) = 0.30(0.08) = \mathbf{0.024}$.
(b) $P(D) = 0.021 + 0.024 = \mathbf{0.045}$ (4.5%).
(c) $P(A \mid D) = \dfrac{0.021}{0.045} = \mathbf{0.4667}$.
(d) $P(B \mid D) = \dfrac{0.024}{0.045} = \mathbf{0.5333}$, and $0.4667 + 0.5333 = 1$. ✓
Machine B makes under a third of the items but accounts for the majority of the defectives.

Q30. *[10 marks]* Apply Bayesian inference. A coin is equally likely to be fair ($p = 0.5$) or biased ($p = 0.75$). It is tossed four times and lands heads every time.

(a) State the two priors.

(b) Compute the likelihood of four heads under each hypothesis.

(c) Compute the unnormalised posterior for each.

(d) Normalise to obtain both posterior probabilities.

Syllabus: Bayesian inference • Prior, likelihood and posterior • CO3 • Bloom L3 (Apply) • Medium • Analytical • Start word: Apply

Solution. **(a)** $P(\text{fair}) = P(\text{biased}) = \mathbf{0.5}$.
(b) $P(HHHH \mid \text{fair}) = 0.5^4 = \dfrac{1}{16} = \mathbf{0.0625}$; $P(HHHH \mid \text{biased}) = 0.75^4 = \dfrac{81}{256} \approx \mathbf{0.3164}$.
(c) Unnormalised: $0.5\left(\frac{1}{16}\right) = \frac{1}{32}$ and $0.5\left(\frac{81}{256}\right) = \frac{81}{512}$. Writing both over 512: $\frac{16}{512}$ and $\frac{81}{512}$.
(d) The total is $\frac{97}{512}$, so

$$P(\text{fair} \mid HHHH) = \frac{16}{97} \approx \mathbf{0.1649}, \qquad P(\text{biased} \mid HHHH) = \frac{81}{97} \approx \mathbf{0.8351}$$

Four heads move the belief from even odds to roughly 5 to 1 in favour of the biased coin.

Q31. *[10 marks]* Evaluate the sample $n = 25$, $\bar{x} = 108$, $s = 15$ against the claim that the population mean is 100, at the 5% level with $t_{0.025,24} = 2.064$.

(a) Compute the standard error.

(b) Compute the t-statistic.

(c) Compare $|t|$ with the critical value and state the decision.

(d) Compute the 95% confidence interval.

Syllabus: Frequentist inference • One-sample t-test • CO3 • Bloom L5 (Evaluate) • Difficult • Analytical • Start word: Evaluate

Solution. **(a)** $\text{SE} = \dfrac{15}{\sqrt{25}} = \dfrac{15}{5} = \mathbf{3.0}$.

(b) $t = \dfrac{108 - 100}{3.0} = \mathbf{2.667}$.

(c) $|t| = 2.667 > 2.064$, so we **reject** H_0 at the 5% level: the evidence against a population mean of 100 is significant.

(d) $108 \pm 2.064(3.0) = 108 \pm 6.192$, giving [**101.808**, **114.192**]. The interval excludes 100, which agrees with the rejection in part (c). ✓

Q32. *[10 marks]* Construct the least-squares regression line for $x = 1, 2, 3, 4, 5$ and $y = 3, 5, 4, 8, 10$.

(a) Compute $\bar{x}$, $\bar{y}$, S_{xy} and S_{xx}.

(b) Compute the slope and the intercept.

(c) Compute the fitted values and the residuals.

(d) Compute R^2.

(e) Predict y at $x = 6$.

Syllabus: Regression analysis • Least-squares line and R-squared • CO3 • Bloom L6 (Create) • Difficult • Analytical • Start word: Construct

Solution. **(a)** $\bar{x} = 3$, $\bar{y} = \dfrac{30}{5} = 6$.

x	y	$x - \bar{x}$	$y - \bar{y}$	product	$(x - \bar{x})^2$
1	3	-2	-3	6	4
2	5	-1	-1	1	1
3	4	0	-2	0	0
4	8	1	2	2	1
5	10	2	4	8	4
			Sum	**17**	**10**

(b) $b = \dfrac{17}{10} = \mathbf{1.7}$; $\quad a = 6 - 1.7(3) = \mathbf{0.9}$, so $\hat{y} = \mathbf{0.9 + 1.7x}$.

(c) Fitted: 2.6, 4.3, 6.0, 7.7, 9.4. Residuals: 0.4, 0.7, -2.0, 0.3, 0.6.

(d) $\text{SSE} = 0.16 + 0.49 + 4.00 + 0.09 + 0.36 = 5.10$ and $S_{yy} = 9 + 1 + 4 + 4 + 16 = 34$, so

$$R^2 = 1 - \frac{5.10}{34} = 1 - 0.15 = \mathbf{0.85}$$

(e) $\hat{y}(6) = 0.9 + 1.7(6) = \mathbf{11.1}$ — an extrapolation beyond the observed range $1 \le x \le 5$.

Unit IV — Graph Theory

Maps to CO4 — analyze different graph theory concepts and use them in solving practical problems.

Syllabus: graph basics, graph types, graph searching algorithms, graph traversing algorithms, graph colouring problem, max-flow min-cut algorithm.

This unit: 8 short-answer questions (2 marks) and 4 long-answer questions (10 marks) = **56 marks**.

Part A — Short Answer Questions — 2 marks each

Q33. *[2 marks]* State the number of edges in a graph whose degree sequence is $(4, 3, 3, 2, 2, 2)$, using the handshaking lemma.

Syllabus: Graph basics • Handshaking lemma • CO4 • Bloom L1 (Remember) • Easy • Analytical • Start word: State

Solution.

$$\sum \deg(v) = 4 + 3 + 3 + 2 + 2 + 2 = 16 = 2E \;\Rightarrow\; E = \mathbf{8}$$

Q34. *[2 marks]* State the number of edges in the complete graph K_7 and in the complete bipartite graph $K_{3,5}$.

Syllabus: Graph types • Complete and complete bipartite graphs • CO4 • Bloom L1 (Remember) • Easy • Analytical • Start word: State

Solution.

$$|E(K_7)| = \frac{7 \times 6}{2} = \mathbf{21}, \qquad |E(K_{3,5})| = 3 \times 5 = \mathbf{15}$$

Q35. *[2 marks]* Illustrate the adjacency matrix of the graph on $\{A, B, C, D\}$ with edges AB, AD, BC, BD, CD. List the degree of every vertex and verify the handshaking lemma.

Syllabus: Graph basics • Adjacency matrix and degrees • CO4 • Bloom L2 (Understand) • Easy • Analytical • Start word: Illustrate

Solution.

	A	B	C	D
A	0	1	0	1
B	1	0	1	1
C	0	1	0	1
D	1	1	1	0

Degrees: $\deg A = 2$, $\deg B = 3$, $\deg C = 2$, $\deg D = 3$.

Sum $= 2 + 3 + 2 + 3 = 10 = 2(5)$, and the graph has **5** edges. ✓

Q36. *[2 marks]* Illustrate the Eulerian condition for the graph on $\{A, B, C, D\}$ with edges AB, AD, BC, BD, CD: count the odd-degree vertices and state whether an Eulerian circuit, an Eulerian path, or neither exists.

Syllabus: Graph types • Eulerian path and circuit • CO4 • Bloom L2 (Understand) • Medium • Analytical • Start word: Illustrate

Solution. Degrees: $A = 2$, $B = 3$, $C = 2$, $D = 3$, so exactly **2** vertices have odd degree (B and D).

The graph is connected with exactly two odd-degree vertices, so an **Eulerian path exists** (it must start at B and end at D, or the reverse), but **no Eulerian circuit** exists.

Q37. *[2 marks]* Determine the BFS order from vertex A for the graph with adjacency lists $A : \{B, C, D\}$, $B : \{A, E\}$, $C : \{A, E\}$, $D : \{A, F\}$, $E : \{B, C, F\}$, $F : \{D, E\}$, breaking ties alphabetically, and state the level of each vertex.

Syllabus: Graph traversing algorithms • Breadth-first search • CO4 • Bloom L3 (Apply) • Medium • Analytical • Start word: Determine

Solution. BFS order: **A**, **B**, **C**, **D**, **E**, **F**.

Levels: $A = 0$; $B = C = D = 1$; $E = F = 2$. (E is first reached from B and F from D.)

Q38. *[2 marks]* Calculate the chromatic number of the cycle C_6, the cycle C_7 and the complete graph K_5.

Syllabus: Graph colouring problem • Chromatic number • CO4 • Bloom L3 (Apply) • Easy • Analytical • Start word: Calculate

Solution.

$$\chi(C_6) = \mathbf{2} \text{ (even cycle)}, \quad \chi(C_7) = \mathbf{3} \text{ (odd cycle)}, \quad \chi(K_5) = \mathbf{5}$$

Every vertex of K_5 is adjacent to all four others, so no two may share a colour.

Q39. *[2 marks]* Compute the shortest distance from A to every other vertex in the weighted directed graph with edges $A \to B = 3$, $A \to C = 7$, $B \to C = 2$, $B \to D = 9$, $C \to D = 1$.

Syllabus: Graph searching algorithms • Shortest path distances • CO4 • Bloom L3 (Apply) • Medium • Analytical • Start word: Compute

Solution.

$$d[A] = 0, \quad d[B] = \mathbf{3}$$

$d[C] = \min(7,\ 3 + 2) = \mathbf{5}$ via $A \to B \to C$.

$d[D] = \min(3 + 9,\ 5 + 1) = \mathbf{6}$ via $A \to B \to C \to D$.

In both cases the direct edge from A loses to the two-step route.

Q40. *[2 marks]* Analyze a flow network in which a cut separates $\{s\}$ from the rest. The edges leaving s are $s \to a$ of capacity 9, $s \to b$ of capacity 6 and $s \to c$ of capacity 4. Compute the capacity of this cut and state the upper bound for the maximum flow.

Syllabus: Max-flow min-cut algorithm • Cut capacity • CO4 • Bloom L4 (Analyse) • Medium • Analytical • Start word: Analyze

Solution.

$$c(S, T) = 9 + 6 + 4 = \mathbf{19}$$

Every unit of flow must cross this cut, so the maximum flow is at most **19**.

Part B — Long Answer Questions — 10 marks each

Q41. *[10 marks]* Apply Dijkstra's algorithm from vertex A to the weighted directed graph with edges $A \to B = 5$, $A \to C = 3$, $B \to D = 2$, $C \to E = 5$, $E \to D = 6$, $D \to F = 8$.

(a) Compute the shortest distance to every vertex.

(b) State the finalisation order.

(c) List the predecessor of each vertex.

(d) State the shortest path from A to F and its cost.

Syllabus: Graph searching algorithms • Dijkstra's algorithm • CO4 • Bloom L3 (Apply) • Medium • Analytical • Start word: Apply

Solution. **(a)**

Vertex	A	B	C	D	E	F
$d[\cdot]$	0	5	3	7	8	15

(b) Finalisation order (by increasing distance): **A(0)**, **C(3)**, **B(5)**, **D(7)**, **E(8)**, **F(15)**.

(c) $\pi[B] = A$, $\pi[C] = A$, $\pi[D] = B$, $\pi[E] = C$, $\pi[F] = D$.

(d) $\mathbf{A \to B \to D \to F}$ with cost $5 + 2 + 8 = \mathbf{15}$. Note that D is reached through B (cost 7), not through E (which would cost $8 + 6 = 14$).

Q42. *[10 marks]* Examine the directed acyclic graph with edges $A \to B$, $A \to C$, $B \to D$, $B \to E$, $C \to E$, $D \to F$, $E \to F$ by depth-first search, visiting vertices alphabetically.

(a) Record the discovery and finish times.

(b) List the vertices by decreasing finish time.

(c) State the topological ordering.

(d) Check each of the seven edges against that ordering.

Syllabus: Graph traversing algorithms • Depth-first search and topological sort • CO4 • Bloom L4 (Analyse) • Difficult • Analytical • Start word: Examine

Solution. **(a)**

Vertex	A	B	C	D	E	F
Discovery	1	2	10	3	7	4
Finish	12	9	11	6	8	5

(b) Decreasing finish time: **A(12)**, **C(11)**, **B(9)**, **E(8)**, **D(6)**, **F(5)**.

(c) Topological ordering: **A**, **C**, **B**, **E**, **D**, **F**.

(d) Positions: $A{=}1, C{=}2, B{=}3, E{=}4, D{=}5, F{=}6$.

Edge	$A \to B$	$A \to C$	$B \to D$	$B \to E$	$C \to E$	$D \to F$	$E \to F$
Positions	$1 < 3$	$1 < 2$	$3 < 5$	$3 < 4$	$2 < 4$	$5 < 6$	$4 < 6$

Every edge runs left to right, so the ordering is valid. ✓

Q43. *[10 marks]* Evaluate the maximum flow in the network with capacities $s \to a = 10$, $s \to b = 4$, $a \to b = 2$, $a \to t = 7$, $b \to t = 6$.

(a) Find augmenting paths and bottlenecks until none remains.

(b) State the maximum flow.

(c) List the vertices reachable from s in the final residual graph.

(d) Compute the cut capacity and compare it with the maximum flow.

Syllabus: Max-flow min-cut algorithm • Ford–Fulkerson and minimum cut • CO4 • Bloom L5 (Evaluate) • Difficult • Analytical • Start word: Evaluate

Solution. **(a)**

Step	Augmenting path	Bottleneck
1	$s \to a \to t$	$\min(10, 7) = \mathbf{7}$
2	$s \to b \to t$	$\min(4, 6) = \mathbf{4}$
3	$s \to a \to b \to t$	$\min(3, 2, 2) = \mathbf{2}$

Final flows: $f(s,a) = 9$, $f(s,b) = 4$, $f(a,b) = 2$, $f(a,t) = 7$, $f(b,t) = 6$.

(b) $|f^*| = 7 + 4 + 2 = \mathbf{13}$. Check at the source: $9 + 4 = 13$. ✓

(c) Residual $c_f(s,a) = 10 - 9 = 1 > 0$, so a is reachable. From a both $a \to t$ and $a \to b$ are saturated, so b and t are not. $\mathbf{S} = \{\mathbf{s}, \mathbf{a}\}$, $\mathbf{T} = \{\mathbf{b}, \mathbf{t}\}$.

(d) $c(S,T) = c(s,b) + c(a,b) + c(a,t) = 4 + 2 + 7 = \mathbf{13}$, equal to the maximum flow, as the max-flow min-cut theorem requires. ✓

Q44. *[10 marks]* Design a conflict-free timetable for five papers A to E, where the pairs sharing students are A–B, A–C, A–D, B–C, C–D, D–E.

(a) List the degree of every vertex.

(b) Arrange the vertices in decreasing degree.

(c) Apply Welsh–Powell and assign a colour to each.

(d) State the number of time slots and give the schedule.

Syllabus: Graph colouring problem • Welsh–Powell colouring • CO4 • Bloom L6 (Create) • Difficult • Analytical • Start word: Design

Solution. **(a)** $\deg A = 3$, $\deg B = 2$, $\deg C = 3$, $\deg D = 3$, $\deg E = 1$. (Sum $= 12 = 2(6)$ for the six conflicting pairs. ✓)

(b) Decreasing degree, ties broken alphabetically: $\mathbf{A}(\mathbf{3})$, $\mathbf{C}(\mathbf{3})$, $\mathbf{D}(\mathbf{3})$, $\mathbf{B}(\mathbf{2})$, $\mathbf{E}(\mathbf{1})$.

(c) Colour 1: take A; C, D and B all conflict with A, but E does not, so colour $1 = \{A, E\}$. Colour 2: take C; D and B both conflict with C, so colour $2 = \{C\}$. Colour 3: take D; B does not conflict with D, so colour $3 = \{B, D\}$.

(d) **3** time slots:

Slot	Papers
1	A, E
2	C
3	B, D

Each of the six conflicting pairs now sits in different slots. ✓

Unit V — Algorithm Complexity Analysis

Maps to CO5 — determine the algorithmic complexity of various mathematical problems.

Syllabus: asymptotic notations, recurrence relations, master theorem, P–NP complete problems, satisfiability problem, searching and sorting problem.

This unit: 8 short-answer questions (2 marks) and 4 long-answer questions (10 marks) = **56 marks**.

Part A — Short Answer Questions — 2 marks each

Q45. *[2 marks]* State the maximum number of comparisons binary search makes on a sorted array of 1024 elements, and give the general expression for n elements.

Syllabus: Searching and sorting problem • Binary search comparisons • CO5 • Bloom L1 (Remember) • Easy • Analytical • Start word: State

Solution.

$$\lfloor \log_2 n \rfloor + 1 = \lfloor \log_2 1024 \rfloor + 1 = 10 + 1 = \mathbf{11}$$

In general the worst case is $\lfloor \log_2 \mathbf{n} \rfloor + \mathbf{1}$ comparisons.

Q46. *[2 marks]* State the worst-case and average-case number of comparisons for linear search on an array of 500 elements when the key is present.

Syllabus: Searching and sorting problem • Linear search comparisons • CO5 • Bloom L1 (Remember) • Easy • Analytical • Start word: State

Solution. Worst case: the key is last, giving **500** comparisons.

$$\text{average} = \frac{500 + 1}{2} = \mathbf{250.5}$$

Q47. *[2 marks]* Illustrate the Master theorem on $T(n) = 3T(n/3) + n$ by computing a, b and $n^{\log_b a}$, and stating the resulting complexity.

Syllabus: Master theorem • Case identification • CO5 • Bloom L2 (Understand) • Easy • Analytical • Start word: Illustrate

Solution. $a = 3$, $b = 3$, so $n^{\log_b a} = n^{\log_3 3} = n^1 = n$.
Here $f(n) = n$ matches $n^{\log_b a}$ exactly, which is **Case 2**:

$$T(n) = \boldsymbol{\Theta}(\mathbf{n} \log \mathbf{n})$$

Q48. *[2 marks]* Illustrate the growth of n^2, $n \log_2 n$ and 2^n by evaluating all three at $n = 16$, and arrange them in increasing order.

Syllabus: Asymptotic notations • Comparing growth rates • CO5 • Bloom L2 (Understand) • Easy • Analytical • Start word: Illustrate

Solution.

$$n \log_2 n = 16(4) = \mathbf{64}, \qquad n^2 = \mathbf{256}, \qquad 2^n = 2^{16} = \mathbf{65\,536}$$

Increasing order: $\mathbf{n \log_2 n} \ < \ \mathbf{n^2} \ < \ \mathbf{2^n}$.

Q49. *[2 marks]* Solve $T(n) = T(n-1) + 3$ with $T(1) = 2$ by iteration, and evaluate $T(40)$.

Syllabus: Recurrence relations • Solution by iteration • CO5 • Bloom L3 (Apply) • Medium • Analytical • Start word: Solve

Solution. Each step adds 3, and there are $n-1$ steps from $T(1)$:

$$T(n) = 2 + 3(n-1) = \mathbf{3n - 1}$$

$$T(40) = 3(40) - 1 = \mathbf{119}$$

Q50. *[2 marks]* Determine the complexity of $T(n) = 9T(n/3) + n^2$ using the Master theorem, stating a, b, $\log_b a$ and the case used.

Syllabus: Master theorem • Application of the Master theorem • CO5 • Bloom L3 (Apply) • Medium • Analytical • Start word: Determine

Solution. $a = 9$, $b = 3$, $\log_3 9 = 2$, so $n^{\log_b a} = n^2$.
$f(n) = n^2 = \Theta(n^2)$ matches, which is **Case 2**:

$$T(n) = \mathbf{\Theta(n^2 \log n)}$$

Q51. *[2 marks]* Calculate, for merge sort on 32 elements, the number of levels in the recursion tree and the worst-case number of comparisons using $n \log_2 n - n + 1$.

Syllabus: Searching and sorting problem • Merge sort • CO5 • Bloom L3 (Apply) • Medium • Analytical • Start word: Calculate

Solution.

$$\text{levels} = \log_2 32 + 1 = 5 + 1 = \mathbf{6}$$

$$\text{comparisons} = 32(5) - 32 + 1 = 160 - 32 + 1 = \mathbf{129}$$

Q52. *[2 marks]* Analyze the subset-sum problem on a set of $n = 25$ numbers. Compute how many subsets an exhaustive search must test, and how many additions are needed to verify one proposed subset.

Syllabus: P–NP complete problems • Verification versus exhaustive search • CO5 • Bloom L4 (Analyse) • Medium • Analytical • Start word: Analyze

Solution.

$$2^{25} = \mathbf{33\,554\,432} \text{ subsets}$$

Verifying one proposed subset needs at most **25** additions.

The gap between 2^{25} and 25 is exactly what separates searching from checking, and is why subset-sum sits in NP.

Part B — Long Answer Questions — 10 marks each

Q53. *[10 marks]* Solve each recurrence.

(a) $T(n) = 3T(n/2) + n^2$.

(b) $T(n) = 8T(n/2) + n^2$.

(c) $T(n) = 2T(n/4) + 1$.

(d) $T(n) = 3T(n-1) + 1$ with $T(0) = 0$; give the closed form and $T(5)$.

Syllabus: Recurrence relations • Master theorem and iteration • CO5 • Bloom L3 (Apply) • Medium • Analytical • Start word: Solve

Solution. **(a)** $a = 3$, $b = 2$, $\log_2 3 \approx 1.585 < 2$, so $f(n) = n^2$ dominates — **Case 3**: $\mathbf{\Theta(n^2)}$.
(b) $a = 8$, $b = 2$, $\log_2 8 = 3 > 2$, so the sub-problems dominate — **Case 1**: $\mathbf{\Theta(n^3)}$.
(c) $a = 2$, $b = 4$, $\log_4 2 = 0.5$, so $n^{\log_b a} = \sqrt{n}$, which beats $f(n) = 1$ — **Case 1**: $\mathbf{\Theta(\sqrt{n})}$.
(d) Iterating: $T(1) = 1$, $T(2) = 4$, $T(3) = 13$, $T(4) = 40$. The closed form is

$$T(n) = \frac{3^n - 1}{2}, \qquad T(5) = \frac{243 - 1}{2} = \mathbf{121}$$

Q54. *[10 marks]* Examine the recurrence $a_n = 5a_{n-1} - 6a_{n-2}$ with $a_0 = 3$ and $a_1 = 8$.

(a) Write the characteristic equation.

(b) Find its roots.

(c) Find A and B from the initial conditions.

(d) Write the closed form and compute a_2, a_3, a_4.

(e) Compute a_2, a_3, a_4 directly from the recurrence and compare.

Syllabus: Recurrence relations • Characteristic equation • CO5 • Bloom L4 (Analyse) • Difficult • Analytical • Start word: Examine

Solution. **(a)** $r^2 - 5r + 6 = 0$.
(b) $(r-2)(r-3) = 0$, so $r = \mathbf{2}$ and $r = \mathbf{3}$.
(c) $a_n = A\,2^n + B\,3^n$ with $A + B = 3$ and $2A + 3B = 8$. Subtracting twice the first from the second gives $B = \mathbf{2}$, hence $A = \mathbf{1}$.
(d) $a_n = \mathbf{2^n + 2 \cdot 3^n}$, so $a_2 = 4 + 18 = \mathbf{22}$, $a_3 = 8 + 54 = \mathbf{62}$, $a_4 = 16 + 162 = \mathbf{178}$.
(e) Directly: $a_2 = 5(8) - 6(3) = 22$; $a_3 = 5(22) - 6(8) = 62$; $a_4 = 5(62) - 6(22) = 178$. The two routes agree. ✓

Q55. *[10 marks]* Evaluate constants showing that $3n^2 + 2n + 5 = \Theta(n^2)$.

(a) Find c_1 with $c_1 n^2 \le 3n^2 + 2n + 5$ for $n \ge 1$.

(b) Find c_2 with $3n^2 + 2n + 5 \le c_2 n^2$ for $n \ge 1$.

(c) State c_1, c_2 and n_0.

(d) Verify at $n = 1$, $n = 2$ and $n = 5$.

Syllabus: Asymptotic notations • Constants for a Theta bound • CO5 • Bloom L5 (Evaluate) • Difficult • Analytical • Start word: Evaluate

Solution. **(a)** For $n \geq 1$ the terms $2n$ and 5 are positive, so $3n^2 + 2n + 5 \geq 3n^2$: take $c_1 = \mathbf{3}$.
(b) For $n \geq 1$, $n \leq n^2$ and $1 \leq n^2$, so $3n^2 + 2n + 5 \leq 3n^2 + 2n^2 + 5n^2 = 10n^2$: take $c_2 = \mathbf{10}$.
(c) $\mathbf{c_1 = 3}$, $\mathbf{c_2 = 10}$, $\mathbf{n_0 = 1}$.
(d)

n	$3n^2$	$3n^2 + 2n + 5$	$10n^2$
1	3	10	10
2	12	21	40
5	75	90	250

The middle column lies between the outer two in every row. ✓

Q56. *[10 marks]* Construct the truth table for the CNF formula $(x_1 \vee \neg x_2) \wedge (x_2 \vee x_3) \wedge (\neg x_1 \vee \neg x_3)$.

(a) State how many rows the table has.

(b) Evaluate each clause for every assignment.

(c) List every satisfying assignment.

(d) State how many assignments satisfy the formula.

Syllabus: Satisfiability problem • Truth table for a CNF formula • CO5 • Bloom L6 (Create) • Difficult • Analytical • Start word: Construct

Solution. **(a)** Three variables give $2^3 = \mathbf{8}$ rows.
(b)

x_1	x_2	x_3	$x_1 \vee \neg x_2$	$x_2 \vee x_3$	$\neg x_1 \vee \neg x_3$	ϕ
F	F	F	T	**F**	T	F
F	F	T	T	T	T	**T**
F	T	F	**F**	T	T	F
F	T	T	**F**	T	T	F
T	F	F	T	**F**	T	F
T	F	T	T	T	**F**	F
T	T	F	T	T	T	**T**
T	T	T	T	T	**F**	F

(c) Satisfying assignments: $(\mathbf{F}, \mathbf{F}, \mathbf{T})$ and $(\mathbf{T}, \mathbf{T}, \mathbf{F})$.
(d) **2** of the 8 assignments, so the formula is **satisfiable**.

Unit VI — Algorithmic and Optimization Approach

Maps to CO6 — discuss different optimization approaches to solve mathematical problems.

Syllabus: greedy approach, dynamic programming, divide and conquer, back tracking, genetic algorithm, nature inspired algorithm and approaches.

This unit: 10 short-answer questions (2 marks) and 4 long-answer questions (10 marks) = **60 marks**.

Part A — Short Answer Questions — 2 marks each

Q57. *[2 marks]* State the value-to-weight ratios for items A (weight 4, value 40), B (weight 8, value 48) and C (weight 10, value 30), and the order a fractional knapsack algorithm uses them.

Syllabus: Greedy approach • Value-to-weight ratios • CO6 • Bloom L1 (Remember) • Easy • Analytical • Start word: State

Solution.

$$r_A = \frac{40}{4} = \mathbf{10}, \qquad r_B = \frac{48}{8} = \mathbf{6}, \qquad r_C = \frac{30}{10} = \mathbf{3}$$

Greedy order (decreasing ratio): **A**, **B**, **C**.

Q58. *[2 marks]* List the activities selected by the greedy activity-selection algorithm from $(1, 3), (3, 6), (2, 7), (6, 8)$, and state how many are selected.

Syllabus: Greedy approach • Activity selection • CO6 • Bloom L1 (Remember) • Easy • Analytical • Start word: List

Solution. The activities are already in increasing order of finish time. Take the first, then each next activity whose start is at least the previous finish:

$$(1, 3) \to (3, 6) \to (6, 8) \to (9, 12)$$

Rejected: $(2, 7)$ starts at $2 < 6$, and $(7, 10)$ starts at $7 < 8$.

4 activities are selected.

Q59. *[2 marks]* Illustrate the greedy coin-change method for an amount of 8 with denominations $\{1, 4, 5\}$: give the greedy solution and the optimal solution, with the number of coins in each.

Syllabus: Greedy approach • Greedy versus optimal coin change • CO6 • Bloom L2 (Understand) • Easy • Analytical • Start word: Illustrate

Solution. **Greedy** takes the largest coin first: $5 + 1 + 1 + 1$, that is **4** coins.

Optimal: $4 + 4$, that is **2** coins.

The greedy rule is not optimal for this coin system, because committing to the 5 leaves a remainder of 3 that only 1s can fill.

Q60. *[2 marks]* Illustrate the saving made by Karatsuba's method by computing the number of sub-multiplications the classical split and Karatsuba each perform after 4 levels of recursion, and computing $\log_2 3$ to four decimal places.

Syllabus: Divide and conquer • Karatsuba multiplication • CO6 • Bloom L2 (Understand) • Medium • Analytical • Start word: Illustrate

Solution. The classical split makes 4 sub-multiplications per level and Karatsuba makes 3, so after 4 levels:

$$\text{classical} = 4^4 = \mathbf{256}, \qquad \text{Karatsuba} = 3^4 = \mathbf{81}$$

That is under a third of the work, and the exponent falls from 2 to $\log_2 3$:

$$\log_2 3 = \mathbf{1.5850}$$

Q61. *[2 marks]* Compute the maximum value for a fractional knapsack of capacity 18 with items A (weight 4, value 40), B (weight 8, value 48) and C (weight 10, value 30).

Syllabus: Greedy approach • Fractional knapsack value • CO6 • Bloom L3 (Apply) • Medium • Analytical • Start word: Compute

Solution. Ratios $10, 6, 3$, so take A, then B, then part of C.

$$A:\ 4\text{ kg},\ 40; \qquad B:\ 8\text{ kg},\ 48; \qquad \text{remaining capacity} = 18 - 12 = 6$$

$$C:\ \frac{6}{10}(30) = 18$$

$$\text{total} = 40 + 48 + 18 = \mathbf{106}$$

Q62. *[2 marks]* Calculate the length of the longest common subsequence of $ABDC$ and $ADBC$, and state one such subsequence.

Syllabus: Dynamic programming • Longest common subsequence length • CO6 • Bloom L3 (Apply) • Medium • Analytical • Start word: Calculate

Solution. The LCS length is **3**.

One such subsequence is **ABC** (positions $1, 2, 4$ of $ABDC$ and $1, 3, 4$ of $ADBC$). ADC is an equally valid answer.

Q63. *[2 marks]* Determine the maximum subarray sum of $[-3, 2, -1, 5, -4, 2]$, and state the subarray that achieves it.

Syllabus: Divide and conquer • Maximum subarray • CO6 • Bloom L3 (Apply) • Medium • Analytical • Start word: Determine

Solution. Running best (Kadane): $-3,\ 2,\ 1,\ \mathbf{6},\ 2,\ 4$.

Maximum $= \mathbf{6}$, from the subarray $[\mathbf{2}, \mathbf{-1}, \mathbf{5}]$.

Q64. *[2 marks]* Apply the 0/1 knapsack method to items of weights $\{2, 3, 5\}$ and values $\{20, 25, 40\}$ with capacity $W = 5$. State the optimal value and the items chosen.

Syllabus: Dynamic programming • 0/1 knapsack • CO6 • Bloom L3 (Apply) • Medium • Analytical • Start word: Apply

Solution. Feasible sets of weight at most 5: $\{1\} = 20$, $\{2\} = 25$, $\{3\} = 40$, $\{1, 2\} = 45$ (weight 5).

Optimal value $= \mathbf{45}$, taking **items 1 and 2** (weight $2 + 3 = 5$, exactly filling the knapsack).

Q65. *[2 marks]* Examine the subset-sum instance $S = \{3, 5, 7, 9\}$ with target $T = 12$. State the number of leaves in the complete binary decision tree, and list every subset whose sum is 12.

Syllabus: Back tracking • Subset sum search tree • CO6 • Bloom L4 (Analyse) • Medium • Analytical • Start word: Examine

Solution. Each element is either taken or not, so the complete tree has $2^4 = \mathbf{16}$ leaves.

Subsets summing to 12: $\{\mathbf{3}, \mathbf{9}\}$ and $\{\mathbf{5}, \mathbf{7}\}$.

Backtracking prunes any branch whose partial sum already exceeds 12, so far fewer than 16 leaves are actually reached.

Q66. *[2 marks]* Analyze the acceptance probability $\exp(-\Delta E/T)$ of simulated annealing for $\Delta E = 3$ at $T = 10$, $T = 3$ and $T = 0.5$, giving each value to four decimal places.

Syllabus: Nature inspired algorithm and approaches • Simulated annealing acceptance probability • CO6 • Bloom L4 (Analyse) • Medium • Analytical • Start word: Analyze

Solution.

$$T = 10: \ e^{-0.3} = \mathbf{0.7408}; \qquad T = 3: \ e^{-1} = \mathbf{0.3679}; \qquad T = 0.5: \ e^{-6} = \mathbf{0.0025}$$

As T falls the same worsening move goes from likely to almost never accepted, which is how cooling turns exploration into exploitation.

Part B — Long Answer Questions — 10 marks each

Q67. *[10 marks]* Apply dynamic programming to the 0/1 knapsack problem with weights $\{2, 3, 5\}$, values $\{20, 25, 50\}$ and capacity $W = 6$.

(a) State the recurrence.

(b) Build the complete 4×7 table.

(c) State the optimal value.

(d) Trace back to identify the items and check their weight and value.

Syllabus: Dynamic programming • 0/1 knapsack table and traceback • CO6 • Bloom L3 (Apply) • Medium • Analytical • Start word: Apply

Solution. **(a)** $V[i, w] = V[i-1, w]$ if $w < w_i$, and $\max(V[i-1, w],\ V[i-1, w-w_i] + v_i)$ if $w \geq w_i$.

(b)

$i \backslash w$	0	1	2	3	4	5	6
0	0	0	0	0	0	0	0
1 $(2, 20)$	0	0	20	20	20	20	20
2 $(3, 25)$	0	0	20	25	25	45	45
3 $(5, 50)$	0	0	20	25	25	**50**	**50**

(c) Optimal value = **50**.

(d) $V[3, 6] = 50 \neq V[2, 6] = 45$, so item 3 is taken; the remaining capacity is $6 - 5 = 1$ and

$V[2,1] = 0$, so nothing further is taken.
Chosen: {item **3**} — weight $\mathbf{5} \le 6$, value **50**. Note that the greedy ratio rule would have taken items 1 and 2 for only 45.

Q68. *[10 marks]* Analyze the longest common subsequence of $X = BADC$ and $Y = ABDC$.

(a) State the recurrence.

(b) Build the complete 5×5 table.

(c) State the LCS length.

(d) Trace back to recover the subsequence.

Syllabus: Dynamic programming • Longest common subsequence with traceback • CO6 • Bloom L4 (Analyse) • Difficult • Analytical • Start word: Analyze

Solution. **(a)** $L[i,j] = L[i-1,j-1] + 1$ if $x_i = y_j$, otherwise $\max(L[i-1,j],\ L[i,j-1])$.
(b)

	$\varnothing$	A	B	D	C
$\varnothing$	0	0	0	0	0
B	0	0	1	1	1
A	0	1	1	1	1
D	0	1	1	2	2
C	0	1	1	2	**3**

(c) LCS length = **3**.
(d) From $L[4,4]$: $C = C$, take C and move to $L[3,3]$; $D = D$, take D and move to $L[2,2]$; $A \neq B$, move to the larger neighbour $L[1,2] = 1$; $B = B$, take B. Reading the taken letters in order gives **BDC**.

Q69. *[10 marks]* Evaluate Huffman coding for the frequencies $A = 3$, $B = 5$, $C = 9$, $D = 14$, $E = 20$.

(a) List the merges in order with the combined weight at each step.

(b) Assign a code to every symbol and state its length.

(c) Compute the total number of encoded bits.

(d) Compute the average code length and the percentage saving against a fixed-length 3-bit code.

Syllabus: Greedy approach • Huffman coding • CO6 • Bloom L5 (Evaluate) • Difficult • Analytical • Start word: Evaluate

Solution. **(a)** $3 + 5 = 8$; $8 + 9 = 17$; $14 + 17 = 31$; $20 + 31 = 51$.
(b)

Symbol	Frequency	Code	Length
E	20	`0`	1
D	14	`10`	2
C	9	`111`	3
A	3	`1100`	4
B	5	`1101`	4

(c) $20(1) + 14(2) + 9(3) + 3(4) + 5(4) = 20 + 28 + 27 + 12 + 20 =$ **107** bits. (Equivalently, the sum of the internal node weights $8 + 17 + 31 + 51 = 107$. ✓)

(d) Total frequency $= 51$, so the average length is $107/51 =$ **2.0980** bits. A fixed-length 3-bit code needs $3(51) = 153$ bits, so the saving is

$$\frac{153 - 107}{153} = \frac{46}{153} = \mathbf{30.07}\%$$

Q70. *[10 marks]* Design one generation of a genetic algorithm maximising $f(x) = x^2$ over 5-bit chromosomes, starting from `10110`, `01001`, `11010`, `00111`.

(a) Decode each chromosome and compute its fitness.

(b) Compute the total and average fitness and each selection probability.

(c) Apply single-point crossover to pair $(1, 2)$ after bit 3 and pair $(3, 4)$ after bit 2.

(d) Compute the new fitness values and the new total and average.

Syllabus: Genetic algorithm • One generation • CO6 • Bloom L6 (Create) • Difficult • Analytical • Start word: Design

Solution. **(a)** and **(b)**

No.	Chromosome	x	$f(x) = x^2$	p_i
1	`10110`	22	484	0.3752
2	`01001`	9	81	0.0628
3	`11010`	26	676	0.5240
4	`00111`	7	49	0.0380
		Total	**1290**	1.000
		Average	**322.5**	

(c) Pair $(1, 2)$ after bit 3: `101|10` and `010|01` give `10101` and `01010`. Pair $(3, 4)$ after bit 2: `11|010` and `00|111` give `11111` and `00010`.

(d)

Chromosome	x	$f(x)$
`10101`	21	441
`01010`	10	100
`11111`	31	961
`00010`	2	4
	Total	**1506**
	Average	**376.5**

The best fitness rises from 676 to **961**, and the average from 322.5 to 376.5.